INCH-POUND
MIL-DTL-85097/2F (OS)
14 APRIL 2003
SUPERSEDING
MIL-DTL-85097/2E (OS)
9 February 2000

DETAIL ASSOCIATED SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) CATAPULT CARTRIDGES, MK 205 MODS 1 AND 2

This associated specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mk 205 Mods 1 and 2 Catapult Cartridges described herein shall consist of this associated specification and the latest issue of MIL-DTL-85097.

1. SCOPE

1.1 Scope. This associated specification covers requirements unique to the Catapult Cartridges, Mk 205 Mods 1 and 2 used in Aircraft Ejection Seat Catapult Mk 19 Mod 0 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified required documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-DTL-85097 USN/USMC PROPELLANT ACTUATED DEVICES (PADs), General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of Department of Defense specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, and drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DOCUMENTS
NAVAL SURFACE WARFARE CENTER (CAGE Code 14083)

511-174TD-004	Propellant, NOSIH-BC-11, NSWC/IH-EC-02, NSWC/IH-EC-04, NSWC/IH-EC-05
510-174TD-004	Propellant, NSWC/IH-EC-08

DRAWINGS
NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL772AS400	Catapult Cartridge, Mk 205 Mods 1 and 2

(Application for copies of documents and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 4230, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

STENCEL AERO ENGINEERING CORPORATION

No. 13T1618 Rev.A (30 September 1985)	Final Report Lot Acceptance Test Data Analysis, Mk 205 Mod 1 Catapult Cartridge

(Application for copies should be addressed to Stencel Aero Engineering , Talley Industries Co, 25401 N Central Ave, Phoenix, AZ 85027-9801)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements in this associated specification, unless in direct conflict with the basic specification, MIL-DTL-85097, shall be considered additional, not substitute. The MK 205 Mods 1 and 2 cartridges shall conform to DL772AS400. If there is a direct conflict with the basic specification, this document shall take precedence.

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3.2 Propellant

3.2.1 Mk 205 Mod 1. The Mk 205 Mod 1 propellant shall conform to the requirements of MIL-DTL-85097 and 511-174TD-004. Testing shall be in accordance with 4.1.1.1.

3.2.2 Mk 205 Mod 2. The Mk 205 Mod 2 propellant shall conform to the requirements of MIL-DTL-85097 and 510-174TD-004. Testing shall be in accordance with 4.1.1.2.

3.3 Propellant aging.

3.3.1 Mk 205 Mod 1. The Mk 205 Mod 1 propellant shall meet the aging requirements of MIL-DTL-85097 and 511-174TD-004. Testing shall be in accordance with 4.1.1.1.

3.3.2 Mk 205 Mod 2. The Mk 205 Mod 2 propellant shall meet the aging requirements of MIL-DTL-85097 and 510-174TD-004. Testing shall be in accordance with 4.1.1.2.

3.4 Ballistic performance. The ballistic performance shall conform to 3.4.1 and 3.4.2. Testing shall be conducted in accordance with 4.1.2.

3.4.1 With Mk 19 Mod 0 catapult. When tested as part of the Mk 19 Mod 0 Catapult, ballistic performance of catapult cartridges Mk 205 Mods 1 and 2 shall conform to the requirements of Table I. Thrust of the Mk 205 shall be recorded for information only when tested as part of the Mk 19 Mod 0 Catapult.

TABLE I. Ballistic performance requirements.

Parameter	315 lb (3%) -65 ± 5°F		315 lb (3% 165 ± 5°F		402 lb (98% -65 ± 5°F		402 lb (98%) 165 ± 5°F	
	Min	Max	Min	Max	Min	Max	Min	Max
Acceleration (g)	-	16.7	-	17.9	-	16.4	-	16.8
Onset (g/s)	-	200	-	243	-	196	-	190
Sled velocity (ft/s)	46	-	50	-	43	-	48	-
Ignition delay (ms)	-	18	-	12	-	18	-	12
Port pressure (psi)	1,100	-	1,450	-	1,500	-	1,900	-
Initiation pressure (psig)	400	650	400	650	400	650	400	650

3.4.2 Without Mk 19 Mod 0 catapult. When tested on the dual nozzle static test fixture without the Mk 19 Mod 0 Catapult, ballistic performance of catapult cartridges Mk 205 Mods 1 and 2 shall conform to the requirements of Tables II and III. Tables II and III were developed from the "Final Report Lot Acceptance Test Data Analysis Mk 205 Mod 1 Catapult Cartridge", Stencel Aero Engineering Corporation Document No. 13T1618, Revision A, 20 September 1985. To determine pressure/thrust values use the best fit line equation which follows.

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The best fit line to a set of n data points [X,Y] is defined as Y' = a+bX, where:

$$a = \frac{\sum x_i^2 \sum y_i - \sum x_i \sum x_i y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$b = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

3.4.3 Chamber pressure. Combustion chamber pressure versus time shall be recorded for preproduction units only.

TABLE II. Alternate ballistic performance requirements.

Parameters	-65 ± 5°F Minimum	-65 ± 5°F Maximum	165 ± 5°F Minimum	165 ± 5°F Maximum
Ignition delay (ms)	-	18	-	12
Initiation pressure (psig)	400	650	400	650
Maximum thrust (lb)	215	350	230	421
Burn time (ms)	230	350	190	250

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TABLE III. Acceptance/rejection thrust limits.

-65°F		165°F	
* Zero time (s)	Minimum thrust (lb)	* Zero time (s)	Maximum thrust (lb)
0.0000	0.0	0.0000	0.0
0.0010	25.0	0.0015	240.0
0.0400	45.0	0.0050	240.0
0.0500	50.0	0.0250	125.0
0.0700	58.0	0.0400	125.0
0.0900	68.0	0.0550	143.0
0.1100	78.0	0.0700	166.0
0.1300	90.0	0.0900	201.0
0.1500	105.0	0.1100	240.0
0.1900	135.0	0.1400	301.0
0.2200	165.0	0.1800	385.0
0.2500	200.0	0.1950	417.0
0.2650	215.0	0.2000	418.0
0.2675	50.0	0.2100	421.0
0.2700	0.0	0.2500	421.0
		0.2510	275.0
		0.2560	15.0
		0.2580	0.0

- Zero time is the time beginning at the first indication of thrust.

3.5 Radiographic inspection. The Mk 205 Mods 1 and 2 Rocket Motor shall have no defects when subjected to radiographic inspection in accordance with 4.1.3.

4. VERIFICATION.

4.1 First article/quality conformance. First article and quality conformance tests shall be performed as specified in MIL-DTL-85097 and shall consist of the inspections and tests listed in 4.1.1 through 4.1.3.

4.1.1 Propellant and propellant aging.

4.1.1.1 Mk 205 Mod 1. The contractor shall submit MK 205 Mod 1 propellant and propellant aging test results in accordance with MIL-DTL-85097 and 511-174TD-004. Units shall meet the requirements of 3.2.1 and 3.3.1.

4.1.1.2 Mk 205 Mod 2. The contractor shall submit MK 205 Mod 2 propellant and propellant aging test results in accordance with MIL-DTL-85097 and 510-174TD-004. Units shall meet the requirements of 3.2.2 and 3.3.2.

4.1.2 Ballistic performance. Ballistic testing of the Mk 205 Mods 1 and 2 shall be performed in accordance with MIL-DTL-85097. Units shall meet the requirements of 3.4.

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4.1.2.1 Conditioning temperatures. The conditioning temperatures and tolerances shall be as follows:

a. Cold: -65 ± 5°F.

b. Hot: 165 ± 5°F.

4.1.2.2 Conditioning period. Time for temperature conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: Not less than 3 hours.

b. Temperature cycling: Not less than 4 hours.

4.1.3 Radiographic inspection. Units shall meet the requirements of 3.5 when subjected to radiographic inspection. All components or assemblies shall be radiographed in two views 90° apart, as specified by each unit's applicable drawings (see 2.2.2).

5. PACKAGING

(This section is not applicable to this specification.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The catapult cartridges Mk 205 Mods 1 and 2 are intended for use in the ejection seat catapult used in the AV/TAV-8 and A-7 aircraft to ensure that the occupant/seat combination is propelled safely out of the aircraft. As the catapult is designed for use only in military aircraft, the cartridges have no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable.

6.3 Definitions.

6.3.1 Ignition delay time. Ignition delay time is the elapsed time from shear pin release to start of thrust.

6.3.2 Zero time. Zero time is the time beginning at the first indication of thrust.

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6.4 Subject term (keyword) listing.

Ballistic performance
Propellant
AV/TAV-8
A-7

6.5 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes

Preparing activity:
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3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) CATAPULT CARTRIDGES, MK 205 MODS 1 AND 2

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-1973 (2) DSN 354-1973
c. ADDRESS (*Include Zip Code*) Documentation Branch (Code 4230F) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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